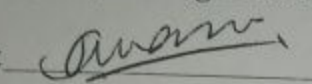


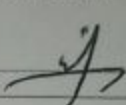


Air University
(End Semester Examination: Spring-2025)

Subject: **Linear Algebra**
Course Code: **MA-201**
Class: **BS-CYS**
Semester: **II**
Section: **A, B (Evening Session)**

Total Marks: **100**
Date: **01/07/2025**
Time:
Duration: **3 Hours**
FM Name: **Mr. Amir Shahzad**

HoD Signature: 

FM Signature: 

IMPORTANT INSTRUCTIONS:

- Attempt all questions
- This examination carries 45% weight toward the final grade
- Scientific calculator is allowed

Q. No. 1 (CLO-3) (PLO-2)		20 Marks
a	Find the least square solutions, the least square error vector, and the least square error for the linear system. $A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & -4 & 3 \\ 1 & 10 & -7 \end{bmatrix}, b = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$	20
Q. No. 2 (CLO-2) (PLO-1)		20 Marks
a*	A real estate company wants to predict the price of a house based on its size (in square feet). Using a dataset of previously sold houses, build a linear regression model to predict house prices. What is the predicted price of a house that is 2,000 square feet, based on your model?	10
b*	Identify the LU factorization of the given matrix A. $A = \begin{bmatrix} 1 & -6 & 3 \\ 5 & -8 & 1 \\ -1 & 7 & 0 \end{bmatrix}$	10
Q. No. 3 (CLO-4) (PLO-2)		20 Marks
a*	A Hill 2-cipher is intercepted that starts with the pairs SL HK Find the deciphering and enciphering matrices, given that the plaintext is known to start with the word ARMY .	20
Q. No. 4 (CLO-5) (PLO-1)		20 Marks
a*	Find the characteristic equation and Diagonalize the following matrix (if possible). $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$	12

b.	Show that the set $\{v_1, v_2, v_3\}$ is an orthogonal basis for R^3. Then express "x" as a linear combination of the v's. $v_1 = \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, u_3 = \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}, x = \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$	8
Q. No. 5 (CLO-1) (PLO-1)		20 Marks
a.	Identify the LU factorization of the given matrix A. $A = \begin{bmatrix} 3 & -6 & 3 \\ 6 & -7 & 2 \\ -1 & 7 & 0 \end{bmatrix}$	10
b.	Find the spanning set for the null space of the matrix $\begin{bmatrix} -3 & 6 & -1 & 1 \\ 1 & -2 & 2 & 3 \\ 2 & -4 & 5 & 8 \end{bmatrix}$	5
c.	How many rows and columns must a matrix A have to define a mapping from R^4 into R^5 by the rule $T(x) = Ax$?	5

*****Best of Luck*****

***** End *****